

OrthoHMM: HMM-Centered Group Inference With Phylogenetic Refinement

Condensed scientific draft, updated 4 October 2026. Not submission-ready. This resource-complete working revision preserves the [3 October text](#) and its rendered reviews; it does not relabel their historical resource statements or change the frozen method, benchmark scores or endpoints. The [extended manuscript](#) retains detailed methods, historical analyses, citations and audit records. This main text does not supersede frozen protocols or historical result manifests.

Abstract

Orthology inference requires balancing homolog recovery against separation of paralogs. We evaluated an HMM-centered pipeline with candidate-family expansion and phylogenetic refinement against established orthology tools. QfO and OrthoBench were the primary development-exposed benchmarks; a frozen YGOB evaluation assessed transfer to additional taxa. Phylogenetic OrthoHMM improved on its high-sensitivity configuration but did not consistently outperform full OrthoFinder. On OrthoBench, their group-recovery F1 values were 74.11% and 72.74%, respectively. On corrected QfO, OrthoFinder had higher VGNC, SwissTrees and TreeFam-A point estimates, whereas OrthoHMM had higher GO/EC similarity and FAS. A matched-recall simulation control supported an initial HMM-search contribution in a fixed graph procedure, not full-pipeline superiority. Family-disjoint generalization and uncertainty for several QfO endpoints remain unresolved. The completed shared-host matched-resource panel retains all 27 attempts, with 24 eligible measurements; it does not establish isolated comparative efficiency. The evidence supports a bounded contribution and an explicit precision-recall trade-off, not universal accuracy or efficiency claims. Synthetic null scoring revealed composition-dependent behavior of the approximate significance filter. Heterogeneous-length simulations favored full OrthoFinder; supplied generating trees did not establish an accuracy gain, and stronger topology perturbations reduced F1 and recall in several conditions.

Methods

The retained configurations are high-sensitivity OrthoHMM and the satellite_v2 phylogenetic pipeline. The original OrthoHMM preprint (Steenwyk et al. 2024) describes its lineage, not all subsequent implementation changes. The prospective method was frozen at 7f3a9e4, with BLOSUM62, E-value threshold 1e-4, Leiden CPM resolution 0.1 and seed 4 (Traag, Waltman, and van Eck 2019; Traag, Van Dooren, and Nesterov 2011). The phylogenetic configuration expands candidate families, infers gene and species trees, and applies positive-paralogy pair inference. Historical runs retain their actual revisions rather than inheriting this prospective pin. The [method diagram](#) distinguishes initial search, profile refinement, candidate expansion and phylogenetic inference.

Initial search uses banded local maximum-path scoring with match, insert and delete states, integer BLOSUM62 emissions and fixed transition costs. Raw scores receive an approximate significance filter before geometric-mean length normalization. MSA-derived profiles learn match emissions, while retaining uniform transition costs; strict expansion requires both a member-score threshold and an initial

sequence-supported anchor. This is not demonstrated HMMER/phmmer equivalence or calibrated assignment confidence. The numerical specification records the recurrence, constants, profile construction and acceptance rules at the full frozen source revision. No scoring/default change accompanies it.

A prespecified null-score protocol examined that significance filter without fitting parameters. Three residue compositions and lengths 50/150/400 each used ten seeds and 1,000 independently drawn query-target pairs per seed. Each pair had a one-target database and was scored with full and width-64 bands: 90,000 independent pairs, not 180,000 independent observations. Five fixed cutoffs yielded 90 tail endpoints with Bonferroni-adjusted exact binomial intervals. These intervals describe the synthetic generator, not reference-family or orthology uncertainty.

Comparators were OrthoFinder 3.1.5 (David M. Emms et al. 2026a, 2026b), SonicParanoid 2.0.9 (Cosentino, Sriswasdi, and Iwasaki 2024), ProteinOrtho 6.3.6 (Klemm, Stadler, and Lechner 2023), FastOMA 0.3.5 (Majidian et al. 2025) and OrthoMCL 1.4 (Li, Stoeckert, and Roos 2003). These method references do not replace run-specific executable provenance. OrthoFinder's sequence-only MCL checkpoint was a diagnostic output, not a separately finalized phylogenetic analysis. FastOMA used a supplied OrthoFinder species tree. QfO inputs included native ortholog pairs, native post-clustering relations or group-derived cross-species pairs as appropriate; these are not interchangeable output semantics. The generated comparison reports each conversion and prediction count.

OrthoBench measures curated group recovery (David M. Emms and Kelly 2020). QfO (Quest for Orthologs consortium et al. 2016; Altenhoff et al. 2020) reports GO and EC similarity, VGNC, SwissTrees and TreeFam-A F1, and FAS separately. Their arithmetic mean is a project-defined secondary summary, not an official QfO F1. Three Kingdoms is supplementary BUSCO-reference recovery (Manni et al. 2021), not genome-wide orthology truth. QfO and OrthoBench were repeatedly inspected during development. The YGOB protocol, using the curated homology/syteny resource (Byrne and Wolfe 2005), froze evaluation before test-score inspection; overlap assessment limits its interpretation to novel-taxon transfer rather than family-disjoint validation.

Uncertainty analyses resample declared reference units and recompute the benchmark statistic within replicates. They retain prespecified multiplicity adjustments, negative findings and unavailable contrasts. Curated families are not automatically independent: shared evolutionary history and predictions joining families can violate exchangeability. Dependent protein pairs are not treated as independent observations to obtain narrower intervals.

The prespecified QfO parameter neighborhood retained seven arms: the frozen control and six one-at-a-time changes. CPM resolutions were 0.08/0.12 versus 0.1, candidate minimum normalization 0.024/0.036 versus 0.03, and minimum margin 1.2/1.8 versus 1.5. SwissTrees F1 was recomputed from macro precision and recall in each of 100,000 shared family-bootstrap draws, not averaged across family F1 values. The seed was 20260925; Bonferroni adjustment retained all 18 planned endpoints across six contrasts and three metrics. This is development-exposed local sensitivity, not independent method selection. Frozen parameter protocol.

Two evolutionary simulation panels used Zombi and Pyvolve (Davín et al. 2019; Spielman and Wilke 2015), ten seeds and seven conditions each: baseline, divergence, duplication/loss turnover, their combination, missing proteins, uneven sampling and a taxon-count control. Event-derived cross-species truth was not replaced by ancestral-family membership. Fixed 300-residue sequences and heterogeneous ancestral-family lengths of 100-500 residues were analyzed separately; descendants retain family length, without within-family indels. Native admission, including finite comparator graph weights, preceded

scoring. Within each panel, 20,000 paired successful-seed resamples recomputed mean seed-level metrics, with a 14-contrast adjustment for primary F1. Failures were not imputed; available-case means from different seed sets are not paired effects. [Simulation protocols and results](#).

On the 70 heterogeneous-length datasets, phylogenetic OrthoHMM and full OrthoFinder also used supplied generating trees and deterministic rooted nearest-neighbor interchanges at clade distances two and four. Generating trees are oracle diagnostics, not achievable end-to-end inference or guaranteed accuracy upper bounds. All 126 F1/precision/recall endpoints were exploratory, with 20,000 paired-seed resamples, seed 20260918 and Bonferroni adjustment. [Tree-control protocol](#).

Shared-Host Resource Measurement

The replacement resource panel uses the local x86 Threadripper host, bizon, not the historical DGX/ARM environment. The frozen panel comprises high-sensitivity OrthoHMM, satellite_v2 OrthoHMM with inferred phylogeny and full OrthoFinder 3.1.5 on nested sets of 4, 8 and 12 proteomes, with three rotated repeats per method and size (27 sequential attempts). The input counts are 73,266, 165,168 and 251,378 proteins. Scientific settings, input bytes and order remain frozen. Native tasks use the same 32-CPU affinity and 128-GiB ceiling; each non-exclusive Slurm allocation reserves 64 slots to accommodate measurement and reporting work. The selected affinity uses distinct physical cores, but allocation does not reserve the host against unrelated processes.

Under the user-authorized 3 October amendment, background analyses do not invalidate a run merely by competing for resources. Launch capacity, source and input identity, process attribution, measurement continuity and failure handling still apply. Process snapshots and pressure observations bracket the native interval and retain background CPU demand, memory pressure and I/O diagnostics. Snapshot coverage is not continuous isolation certification, and CPU pressure can exceed nominal bounds because of non-atomic accounting; pressure magnitudes are diagnostic rather than timing-eligibility thresholds.

The primary endpoints are native-command monotonic elapsed time, native task-subtree CPU-stat change including wrapper work, and native-step lifetime memory peak including the launcher. They exclude separate preparation, conversion and scoring; the peak is not pure algorithm RSS. Terminal scheduler, runtime, environment, resource and native-output evidence is reviewed before the next identity. All attempts and failures are retained without fastest-run selection, automatic retries or estimated overhead subtraction. Median/range summaries require all three eligible repeats; observed ranges are not confidence intervals. Shared-host distortion is unknown and potentially method dependent. Historical DGX and earlier shared-host timings are not pooled with this panel. [Execution amendment](#), [endpoint and continuation contract](#).

Results

Accuracy Depends On The Endpoint

Phylogenetic OrthoHMM achieved 74.1061% OrthoBench F1 versus 72.7365% for full OrthoFinder. In the exploratory eight-method comparison, the F1 difference was +1.370 percentage points (21-endpoint adjusted interval [-7.281, 12.166]). The precision difference was +15.705 [1.175, 30.634] and recall difference -13.151 [-26.947, -0.609]. This is a precision-recall tradeoff, not established F1 superiority. The complete panel uses 100,000 paired RefOG draws and remains conditional on family exchangeability and development exposure. [All 21 contrasts](#). The [complete interval figure](#) shows F1, precision and recall

on the same difference scale without ranking methods by their observed effects. A fresh installed full run reproduced all 59,770 OrthoHMM groups exactly; this establishes reproducibility, not independent accuracy or a controlled runtime comparison. Full-run verification. A second full run using a separately reconstructed base Python and fresh inference/reader environments also reproduced all groups and all 70 family score records. Native pair, confidence, reconciliation-event and hierarchy files were byte-identical to the earlier run. This extends same-host environment reproduction, not biological validation or complete archive restoration. Reconstructed-base verification. A third full run used restored local execution assets and separately acquired upstream inputs. Its independent scientific admission reproduced all 59,770 groups, all 70 family score records and four native TSVs exactly, without reusing phylogenetic checkpoints. This validates local archive-to-results reproduction, not cross-host/OS restoration, independent accuracy or controlled timing. Restored-archive verification.

On corrected QfO, phylogenetic OrthoHMM versus full OrthoFinder scored 0.901690 versus 0.988546 on VGNC, 0.833513 versus 0.848413 on SwissTrees, and 0.614864 versus 0.791918 on TreeFam-A. OrthoHMM had higher GO similarity (0.490349 versus 0.469548), EC similarity (0.965650 versus 0.936130) and FAS (0.762993 versus 0.691422). These endpoints differ in reference scope and eligibility; they do not form a single accuracy ranking. All eight methods.

An exploratory VGNC deletion diagnostic removed each of 16,844 reference blocks and its incident scored pairs in turn, across all eight methods. Every comparator retained a negative F1 difference against full OrthoFinder for every deletion. For phylogenetic OrthoHMM the range was [-8.7353, -8.6729] percentage points. These are sensitivity ranges, not confidence intervals: dependent deletions change the fixed scored table, and single-block sign stability does not establish population or joint-deletion robustness. Native inference and eligibility were not rerun.

For corrected SwissTrees, phylogenetic-minus-full-OrthoFinder F1 was -1.4900 percentage points, with a multiplicity-adjusted interval of [-8.7078, 7.2090]. Phylogenetic-minus-high-sensitivity OrthoHMM was +14.8015 points [4.9791, 26.6015]. These conditional estimates use 18 development-exposed families and a 24-endpoint adjustment; inclusion of zero is not equivalence. Candidate expansion also changes between OrthoHMM configurations, so their contrast does not isolate reconciliation. Protocol and contrast evidence.

All 56 method-pair GO/EC comparisons had identical six-decimal scores on shared scored pairs. Aggregate differences arose from different eligible pair sets and their denominators. Restricting evaluation to intersections would change the endpoint rather than resolve uncertainty. Historical raw hashes were verified for all 16 input tables. Scored-pair audit. The complete pair-composition figure displays the shared-denominator and exclusive-pair terms for all seven comparators against full OrthoFinder. Phylogenetic OrthoHMM's shared pairs account for 90.12% of its GO scored set but 46.40% of OrthoFinder's, and 86.59% versus 58.00% for EC. The original rounded-mean differences, +2.080130 and +2.952029 score points, are the sums of much larger opposing terms. These fractions describe eligible scored pairs, not proteome-wide coverage; the decomposition is arithmetic, not causal attribution or paired uncertainty.

Component Evidence Is Bounded

The corrected QfO factorial supported candidate-expansion-by-reconciliation interactions on SwissTrees F1, while all four profile-refinement F1 intervals included zero. Reconciliation increased precision and reduced recall. These results support conditional component effects, not proof that profile expansion universally improves orthology inference. Complete factorial.

In 35 matched-recall simulation datasets spanning seven conditions and five seeds, HMM-derived search evidence yielded mean downstream graph F1 of 83.6097%, versus 80.5180% for DIAMOND (Buchfink, Reuter, and Drost 2021). The adjusted paired seed-block interval for the +3.0917-point difference was [1.6782, 4.5313]. Profile expansion and phylogeny were off. This is a development-exposed fixed-graph comparison, not a comparison against OrthoFinder or a matched-effort result. Score rankings and hit identities differ, so the design does not isolate a causal mechanism. [Matched-recall control](#).

Synthetic Null Tails Depend On Composition

At the frozen $E < 1e-4$ cutoff, width-64 scoring passed 0/1/0 of 10,000 ordinary-background pairs at lengths 50/150/400. These sparse counts do not establish rare-tail calibration. Half-background-plus-half-glutamine sampling passed 89.83%, 100% and 100%; its adjusted intervals were [88.7484%, 90.8453%] and [99.9181%, 100%] for each longer length. All [90 endpoints and their figure](#) are retained. A hit here is only a forced pair passing the approximate significance gate, not a predicted ortholog or an observed pipeline false positive. No prefilter or biological inference ran. The strong synthetic composition is not a measured real-proteome prevalence. The [audit](#) reproduces all endpoint counts and 180 sparse reference-Python scores, not all native scores. No coefficients, thresholds or defaults were fitted or promoted.

Evolutionary Simulations Retain Failures And Tree Sensitivity

In the fixed-length stress panel, high-sensitivity and phylogenetic OrthoHMM had 70 and 64 admitted datasets out of 70. No OrthoFinder output passed its native-completion/finite-graph gate; all 14 planned comparisons are unavailable, not wins for OrthoHMM. The heterogeneous-length panel admitted 70 and 67 OrthoHMM datasets and 65 full OrthoFinder datasets. Full OrthoFinder had higher paired mean F1 than both OrthoHMM modes in all seven conditions. Adjusted F1 intervals were below zero for all seven high-sensitivity contrasts and four phylogenetic contrasts. Phylogenetic-minus-full-OrthoFinder effects were -11.7433 points for divergence (five paired seeds) and -12.2842 for combined divergence/turnover (eight). These are success-conditioned comparisons, not failure-adjusted population estimates. Species-tree inference failures for OrthoHMM and nonfinite graph weights for OrthoFinder remain explicit. [Fixed stress results](#), [heterogeneous-length results](#).

The complete tree-control analysis retained 560 arm outcomes: 537 scored and 23 failed. No generating-minus-inferred adjusted interval excluded zero. The stronger perturbation reduced F1 and recall with adjusted intervals below zero in four OrthoHMM conditions and two OrthoFinder conditions: 12 endpoints; no precision interval excluded zero. Retained upstream artifacts agreed in 400 comparisons, differed in two and were unavailable in 18. The differing OrthoFinder cases prevent a strict tree-only causal attribution there. These deterministic perturbations and ten planned seeds do not establish robustness to arbitrary trees or empirical posterior uncertainty. [Complete tree results and figure](#).

Local Parameter Sensitivity Did Not Establish Improvement

The [complete seven-arm panel](#) includes separately admitted private-runtime high-CPM recovery. Its SwissTrees F1 was 83.253016% versus 83.351322% for the frozen control, a difference of -0.098306 percentage points with adjusted interval [-1.050447, 0.747525]. All 18 adjusted intervals include zero; no statistically supported parameter improvement is established. Identical observed normalization-arm statistics do not imply identical whole-proteome predictions or equivalence on unseen families. No defaults were changed. The [all-arm figure](#) retains every contrast and the full adjustment denominator.

Historical SIGSEGV, allocator and admission failures remain failed. Recovery followed a validated content-equivalent private control and has separate inference, conversion and score evidence; it neither repairs those earlier attempts nor proves their cause or memory safety. This panel is not an OrthoFinder superiority test or a controlled runtime comparison.

Transfer And Biological Recovery Reveal Trade-Offs

Frozen YGOB group-recovery F1 was 92.233654% for phylogenetic OrthoHMM and 92.318524% for full OrthoFinder. Their difference was -0.084870 points, with adjusted interval [-0.622528, 0.445222]. OrthoHMM had higher precision and lower recall. The evaluation projects predictions onto the reference universe and assumes exchangeable pillars; it does not establish unrestricted generalization or resolved pairwise orthology. YGOB results.

A descriptive partition by the frozen overlap screen retained all original false-positive allocations. In screen-negative pillars, phylogenetic OrthoHMM versus full OrthoFinder had F1 of 57.93% versus 45.02%, precision of 55.92% versus 30.74%, and recall of 60.10% versus 84.03%. In screen-positive pillars, their F1 values were 93.37% versus 95.01%. The negative stratum contained 2,893 singleton pillars out of 3,298, compared with 2,017 out of 6,952 in the positive stratum. This composition difference and the screen's inability to exclude remote homology prevent independent-family or causal interpretations. The all-method stratum figure is descriptive, with no new uncertainty estimate or method tuning.

The prespecified whole-genome-duplicate application used experimental evidence from Kuzmin and colleagues (Kuzmin et al. 2020). All 240 experimental pairs were retained, with 239 input-eligible and 231 shared-reference-pillar pairs. On this development-exposed application, phylogenetic OrthoHMM separated 238 of 239 input-eligible pairs, versus 58 for high sensitivity and 236 for full OrthoFinder. Among the 231 reference-eligible pairs, however, only 193 OrthoHMM separations retained at least one non-*S. cerevisiae* reference homolog with each anchor, versus 56 for high sensitivity and 227 for full OrthoFinder. Mean per-pair homolog coverage was 82.338%, 99.149% and 98.413%, respectively. Coverage counts reference homologs in the union of the anchor groups and can be high even when the anchors are merged; it is not orthology recall. The complete five-method comparison also retains SonicParanoid and the diagnostic OrthoFinder MCL checkpoint.

Phylogenetic OrthoHMM minus full OrthoFinder had a supported-separation difference of -14.719 percentage points, with adjusted interval [-21.645, -8.225], and a coverage difference of -16.075 points [-19.755, -12.496]. These exploratory percentile intervals use 20,000 paired pillar resamples and a 12-endpoint adjustment. Each eligible pair occupies a distinct pillar; these conditional estimates do not establish independent generalization. Supported separation does not establish cross-species copy-specific orthology. In the six prespecified case traces, five focal homologs left their anchor groups during root-lineage reconstruction, before satellite constraints. They remained in other output groups; no example was replaced to improve the result.

A subsequent fixed-tree diagnostic evaluated all four existing root-duplication rules on the same seven candidate families and six examples, with its protocol frozen before alternative outcomes were examined. The supported-children and confidence rules reproduced the species-overlap baseline partitions exactly. The mapped-event rule reduced homolog coverage in four of five reference-eligible examples, without improving supported separation. None recovered the five focal homologs into either anchor group. Native pair predictions and confidence annotations remained unchanged. This post hoc intervention rejects these three alternatives as a repair on the fixed inputs, not as methods in general. It neither establishes topology error nor identifies ancestral-copy truth, and no defaults were changed.

Shared-Host Resource Panel

The snapshot retains 27 of 27 reviewed attempts, 25 with measured native resources and 24 eligible observations. 6 method/size cells have three eligible repeats.

These are shared-host matched-resource observations with 32 native CPUs and a 128-GiB limit per run, not isolated comparative timing. Contention distortion is unknown and potentially method dependent. No background overhead is subtracted.

Tables report median [minimum, maximum] only for cells with three eligible repeats. An unavailable summary is not zero or a failed native inference. Ranges are not confidence intervals. Eligibility counts retain measured exclusions and pre-native aborts rather than selecting the fastest attempts.

Native command wall (seconds)

Method	Proteomes	Eligible/planned	Median [minimum, maximum]
OrthoHMM high sensitivity	4	2/3	Unavailable
OrthoHMM high sensitivity	8	3/3	1238.9352 [1221.4444, 1303.6330]
OrthoHMM high sensitivity	12	3/3	2720.0774 [2621.4976, 3027.0605]
OrthoHMM inferred phylogeny	4	2/3	Unavailable
OrthoHMM inferred phylogeny	8	3/3	1999.5508 [1888.8847, 2086.9473]
OrthoHMM inferred phylogeny	12	3/3	4392.8716 [4042.4262, 4809.2924]
OrthoFinder 3.1.5 full	4	3/3	441.2890 [420.7400, 474.2999]
OrthoFinder 3.1.5 full	8	3/3	1139.9398 [1120.2548, 1179.9703]
OrthoFinder 3.1.5 full	12	2/3	Unavailable

Task-subtree CPU bracket (CPU-seconds)

Method	Proteomes	Eligible/planned	Median [minimum, maximum]
OrthoHMM high sensitivity	4	2/3	Unavailable
OrthoHMM high sensitivity	8	3/3	34395.7481 [33878.9946, 36610.5696]

Method	Proteomes	Eligible/planned	Median [minimum, maximum]
OrthoHMM high sensitivity	12	3/3	78335.8306 [75126.8588, 86415.3192]
OrthoHMM inferred phylogeny	4	2/3	Unavailable
OrthoHMM inferred phylogeny	8	3/3	56008.3471 [52911.7819, 58240.7234]
OrthoHMM inferred phylogeny	12	3/3	125264.7375 [116000.9964, 136293.3322]
OrthoFinder 3.1.5 full	4	3/3	4602.7585 [4455.3737, 5008.8970]
OrthoFinder 3.1.5 full	8	3/3	17734.4088 [17232.2843, 18787.3118]
OrthoFinder 3.1.5 full	12	2/3	Unavailable

Native-step lifetime peak (GiB)

Method	Proteomes	Eligible/planned	Median [minimum, maximum]
OrthoHMM high sensitivity	4	2/3	Unavailable
OrthoHMM high sensitivity	8	3/3	7.724 [7.715, 7.932]
OrthoHMM high sensitivity	12	3/3	10.555 [10.533, 10.730]
OrthoHMM inferred phylogeny	4	2/3	Unavailable
OrthoHMM inferred phylogeny	8	3/3	7.681 [7.677, 7.702]
OrthoHMM inferred phylogeny	12	3/3	10.511 [10.503, 10.603]
OrthoFinder 3.1.5 full	4	3/3	6.775 [6.769, 6.928]
OrthoFinder 3.1.5 full	8	3/3	10.226 [10.215, 10.233]
OrthoFinder 3.1.5 full	12	2/3	Unavailable

Excluded attempt indices: 0, 17, 20. Pre-native abort indices: 17, 20; these have no native resource endpoints, not zero measurements. Unreviewed indices: none.

Across measured attempts, maximum observed foreign CPU demand ranges from 42.1424 to 72.8708 core equivalents. These are process-interval observations, not reservations or estimates of causal slowdown.

CPU includes the native task-subtree wrapper bracket. Peak memory includes the native-step launcher and is not pure algorithm RSS. Preparation, conversion and scoring are outside the native command timer. This single nested taxon series co-varies proteome count and taxon composition, not taxon-invariant

scaling. Native output validation does not provide new prediction-accuracy evidence. Historical DGX and earlier shared-host timings are not pooled.

The first high-sensitivity attempt completed native inference but failed the original process-monitoring cadence criterion. It remains a measured exclusion; the prospectively tested scheduling repair does not convert it to a pass. Two further attempts aborted before native inference: an OrthoFinder/twelve-proteome repeat encountered a preparation hash race, and a phylogenetic OrthoHMM/four-proteome repeat exceeded the environmental-response deadline. Both retain null native endpoints and separately reviewed infrastructure repairs. None of these failures was retried to fill a cell or obtain faster timing. All 27 planned identities are terminal-reviewed, including exclusions; this is not 27 eligible measurements. The three incomplete cells have two eligible repeats each, not completed three-repeat summaries.

[Machine-readable final table](#), [all attempts](#), [complete-cell summaries](#), [final resource figure](#), [final terminal review](#), [generated section and provenance](#).

Discussion And Limitations

The supported contribution is an HMM-centered alternative with measurable component effects and explicit accuracy trade-offs. Initial HMM search has bounded support in matched-recall simulations; additional profile refinement has not demonstrated a general benefit. Neither simulation evidence nor an OrthoBench point advantage establishes superiority over full OrthoFinder.

The approximate significance formula is not universally calibrated under the tested synthetic conditions. The null experiment does not measure real-data orthology false-positive rates or causally explain any benchmark difference; independent calibration would be needed before claiming statistical significance for arbitrary compositions, search spaces, profiles or band settings.

Original TreeFam-A family mappings and complete source trees remain unavailable. Public QfO container searches now cover recognized benchmark build contexts for [all 40 listed tag names](#) at retained digests, without finding original NHX/mapping filename candidates. Other layers and archive-embedded or differently named contents remain outside that search; no family labels are inferred from its negative result. Public archive recovery yielded historical Selectome subtrees, but all are restricted to Euteleostomi. A [taxonomic coverage audit](#) found that 55,933 of 79,320 retained reference relations (70.52%) have at least one endpoint outside that clade. Those relations cannot be reconstructed from these subtrees under a species-consistent mapping; the other 23,387 relations are only potentially in scope, not demonstrated reconstructions. This is a coverage exclusion, not an accuracy effect or a basis for family-level intervals. VGNC dependence and rare-error diagnostics do not justify the candidate confidence-interval procedure. Valid paired uncertainty for GO/EC, FAS and the secondary mean is also unfinished. These gaps must remain visible alongside point estimates. The [corrected FAS sample audit](#) reproduces all eight means, but scored fractions range from 0.0067% to 58.46% of reported eligible pairs and every sample reuses proteins across pairs. Native pair-level standard errors do not resolve this comparison uncertainty. The [requested-score audit](#) found 1-49 missing new scores per method except OrthoMCL, which lacked 1,252 of 9,000. Assuming omitted scores lie in [0,1], its intended-sample mean is bounded by 0.724872-0.736975; this is not a confidence interval. A [native-complexity audit](#) found no above-limit protein in saved new-score pairs, but found them in precomputed pairs. This agrees with the demonstrated cutoff mechanism without identifying historical omitted pairs or establishing comparison bias. The [completed eligible-population audit](#) matches all eight methods' native logged counts, reusing six fixed recounts and freshly scanning two. Pairs absent from the precomputed lookup constitute 5.13% of satellite_v2's eligible set, 16.86% of full OrthoFinder's and 82.05% of its sequence-only checkpoint's. Assuming their scores lie in [0,1] gives conservative full-mean bounds of 0.733966-0.785282 for

satellite_v2 and 0.605946-0.774511 for full OrthoFinder. These overlapping bounds neither establish a full-eligible-set advantage nor replace native sample scores or confidence intervals. Some lookup-missing pairs were newly scored in native samples; those values are not used to tighten these bounds. An [eight-method stratum-weight decomposition](#) separates count rounding from omission-induced changes in the saved mixture. Holding the observed stratum means fixed, native-minus-reweighted-saved-strata differences are +0.001318947 FAS units for OrthoMCL and +0.000030716 for phylogenetic OrthoHMM. These are numerical diagnostics, not corrected benchmark scores or estimates of population bias. Omissions can also change the stratum means; the unknown omitted pair identities, unseeded selection and dependence remain unresolved. No FAS confidence interval or new method ranking follows. The completed simulations and tree perturbations do not cover arbitrary evolutionary conditions, and novel-taxon YGOB testing retains homolog-family overlap with development data.

Recovered OrthoMCL results retain sequence-specific BLAST failures. The separately recovered high-CPM result does not replace failed historical runs; no missing score was imputed and no failure was recategorized as a success. Shared-host runs and the historical DGX panel are descriptive resource records. The replacement matched-resource Threadripper panel has completed under the shared-host amendment, not as an isolation-controlled experiment. Competing workloads may affect tools differently, and identical resource limits do not identify isolated speed or causal speedups. The final figure retains excluded measurements and missing eligible repeats; it cannot support an isolated efficiency ranking. Six cells have three eligible repeats, while three cells retain unavailable summaries. This single nested taxon series does not isolate taxon-count scaling from proteome composition. Whole-study executable/versioned archival packaging remains open, alongside redistribution limitations and journal-specific formatting. Completed resource reporting does not resolve the scientific uncertainty and validation gaps above.

Reproducibility And Availability

The extended manuscript links frozen protocols, source revisions, seeds, input/output manifests, conversion audits, executable workflows and generated figures. Installed inference and independent readback have been validated on full OrthoBench, but that is not proof of all-method cross-host portability. Historical locks are retained as provenance; patched replacement environments must be distinguished from the binaries used for original scores. The [QfO parameter numerical component](#) passed fresh local archive extraction, copied byte verification and reproduction of all 18 endpoints within 1e-12 using unchanged arithmetic. A Python-event guard rejected its original-checkout canary and observed no later original-path events or project-module imports. This is not OS containment, a new independent statistical implementation, raw-reference recount, native inference restoration, cross-host validation or redistribution clearance.

A separate [descriptive-table component](#) restored under standard-library Python 3.12 and reproduced all 208 rows and 984 score/difference cells across four SwissTrees tables and eight methods. This verifies arithmetic from retained sufficient statistics and annotations, not raw-source admission, new benchmark scoring or bootstrap uncertainty. An [input-only private archive workflow](#) then restored the complete duplication and fragment dependency panels, retaining all 11 and 1,774 original record occurrences respectively. Independently pinned archive and binding digests preserve source identities at new locations. All 27 affected regression/options cases passed using the restored inputs, including four raw-export regressions with unchanged assertions. The fresh children blocked original-checkout reads and subprocesses; these Python-event guards are not OS containment. Native annotation extraction/admission was not rerun, and private archives remain unuploaded with redistribution uncleared. Neither component establishes complete executable study restoration or all-method cross-host portability.

The null-score observations retain all 180,000 synthetic score evaluations and their source/runtime pins. Its public numerical readback checks all 90 tail endpoints without loading the native kernel; this is not a complete inference runtime archive.

The standalone YGOB arithmetic replay uses identifier-free retained counts to reproduce all 12 point metrics and six nominal/adjusted interval endpoints after local archive restoration. This is not native re-admission, proof of pillar exchangeability or independent family validation; original transfer scores and overlap limitations remain.

The standalone simulation replay reproduces retained-count arithmetic for the two length panels without historical file reads. It preserves fixed-length unavailable comparisons and variable-length conditional effects; native admission and tree-control inference are not rerun.

The final resource-reporting archive restores outside the repository and reproduces all three table files, PNG pixels and resource-section text exactly from its copied isolated reader. It includes all attempts and failures without opening original evidence paths. This is reporting portability, not raw-accounting or native reproduction, isolated timing, complete study restoration or redistribution clearance. The source/review export integration adds explicit result-helper and final-manuscript selection; fixture tests do not establish an actual full-study archive.

The progress ledger records completed work and unmet requirements. No submission-ready release or archival DOI is claimed. Dated main-text HTML/PDF and review archives preserve the exact inputs of their own snapshots; older versions are not rendered copies of revised Markdown. A refreshed review is not a complete executable study release. This revision incorporates the completed shared-host resource panel while preserving scientific limitations. Its render/review and the executable/versioned archive require their own source-bound validation; older PDF snapshots remain unchanged. No new accuracy inference or endpoint selection accompanies it.

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The bibliography attributes the methods and resources cited above. Full execution evidence and additional dependency references remain in the extended manuscript; this selected list is not complete software attribution.

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Figure Appendix - Working Review

Main text: pages 1-12. Original vector pages and dimensions are preserved. Shared-host timings have unknown contention effects; no isolated speed ranking or publication-readiness claim is made.

A1. HMM-centered workflow

Initial search, profile expansion, candidate families and phylogenetic refinement have distinct outputs. This schematic describes the frozen workflow, not proof of an HMM accuracy advantage.

Source: [benchmark_tools/results/figures_publication_method_20260916/publication_method.pdf](#)

A2. Current eight-method accuracy overview

OrthoBench precision/recall and Three Kingdoms pair F1 use different reference scopes. These descriptive comparisons do not form a cross-benchmark ranking; Three Kingdoms is supplementary conserved-family evidence.

Source: [benchmark_tools/results/figures_current_accuracy_overview_20261002_v2/current_accuracy_overview.pdf](#)

A3. Individual corrected QfO endpoints

GO and EC are similarity scores; VGNC, SwissTrees and TreeFam-A are F1 endpoints, and FAS measures feature-architecture similarity. Eligibility and denominators differ; this plot is descriptive, not paired uncertainty.

Source: [benchmark_tools/results/figures_corrected_qfo_endpoints_20260926/corrected_qfo_endpoints.pdf](#)

A4. OrthoBench paired differences

All 21 method-metric contrasts retain their multiplicity-adjusted RefOG intervals for F1, precision and recall. Development exposure and family-exchangeability assumptions limit inference. An interval containing zero establishes neither superiority nor equivalence.

Source: [benchmark_tools/results/figures_ob_complete_uncertainty_20260928/ob_complete_uncertainty.pdf](#)

Figure Appendix - Working Review

Main text: pages 1-12. Original vector pages and dimensions are preserved. Shared-host timings have unknown contention effects; no isolated speed ranking or publication-readiness claim is made.

A5. HMM, expansion and reconciliation factorial

The corrected SwissTrees factorial retains all cells and component contrasts. Candidate expansion interacts with reconciliation; profile-refinement F1 intervals include zero. These conditional effects do not establish universal improvement.

Source: [benchmark_tools/results/qfo_corrected_factorial_figures_20260919/qfo_factorial_swiss.pdf](#)

A6. GO and EC scored-pair composition

Shared-denominator and exclusive-pair terms reproduce the aggregate differences for all seven comparators against full OrthoFinder. This is arithmetic decomposition of eligible scored sets, not proteome-wide coverage, causal attribution or confidence intervals.

Source: [benchmark_tools/results/figures_qfo_scored_pair_decomposition_20260930/qfo_pair_decomposition.pdf](#)

A7. Matched-recall HMM search control

The fixed-graph diagnostic compares HMM-derived search evidence with DIAMOND on 35 simulation datasets. It is development-exposed, without profile expansion or phylogeny, and does not match computational effort or compare full OrthoFinder.

Source: [benchmark_tools/results/matched_graph_figure_v2_20260926/matched_graph.pdf](#)

A8. Synthetic null-score tails

Ninety endpoints arise from 90,000 independent sequence pairs and 180,000 dependent full/banded evaluations. Composition and length change approximate tail behavior. This is not calibrated orthology confidence, a biological false-positive rate or fitted parameters.

Source: [benchmark_tools/results/figures_frozen_null_scores_20261002_v2/frozen_null_scores.pdf](#)

Figure Appendix - Working Review

Main text: pages 1-12. Original vector pages and dimensions are preserved. Shared-host timings have unknown contention effects; no isolated speed ranking or publication-readiness claim is made.

A9. Fixed-length evolutionary stress

Native failures remain in the fixed-length panel. No OrthoFinder output passes the stated native-completion/finite-graph gate, so its 14 planned comparisons are unavailable rather than OrthoHMM wins.

Source: [benchmark_tools/results/figures_simulation_fixed_native_20260916/simulation_evidence.pdf](#)

A10. Heterogeneous-length evolutionary stress

The heterogeneous-length panel retains admitted and failed runs. Full OrthoFinder has higher paired mean F1 than both OrthoHMM modes in all seven conditions. Effects are conditioned on paired success, not failure-adjusted population estimates.

Source: [benchmark_tools/results/figures_simulation_variable_native_v2_20260916/simulation_evidence.pdf](#)

A11. Species-tree controls and perturbations

Generating and perturbed trees are diagnostic interventions, not independent inferred pipelines or empirical tree posteriors. All 560 outcomes are retained. Two upstream OrthoFinder discrepancies limit strict tree-only causal attribution.

Source: [benchmark_tools/results/figures_simulation_tree_robustness_20260917/simulation_tree_robustness.pdf](#)

A12. Complete local parameter neighborhood

All seven arms, including separately admitted private-runtime recovery, remain in the 18-endpoint adjustment denominator. Every adjusted interval includes zero; no default or scientific setting was promoted.

Source: [benchmark_tools/results/qfo_parameter_complete_export_20261001/qfo_parameter_neighborhood.pdf](#)

Figure Appendix - Working Review

Main text: pages 1-12. Original vector pages and dimensions are preserved. Shared-host timings have unknown contention effects; no isolated speed ranking or publication-readiness claim is made.

A13. YGOB transfer and overlap strata

The frozen separate-clade test supplies bounded transfer evidence. Its subsequent overlap partition is descriptive, preserves false-positive allocations and does not establish family-disjoint confirmation or a second independent test.

Source: [benchmark_tools/results/figures_ygob_overlap_20260928/ygob_overlap_strata.pdf](#)

A14. Whole-genome-duplicate biological application

Anchor separation is distinguished from supported separation and reference-homolog coverage. Coverage can be high for merged anchors and is not orthology recall. Prespecified losses and negative OrthoHMM comparisons are retained.

Source: [benchmark_tools/results/figures_wgd_application_20260917/wgd_application.pdf](#)

A15. All-method OrthoBench error strata

Fourteen frozen descriptive strata retain all eight methods, precision/recall/F1 and empty bins. Development exposure and small bins limit interpretation. Length, copy number and composition are descriptors, not validated fragments, duplication histories or causal error mechanisms.

Source: [benchmark_tools/results/ob_complete_strata_figure_20260928/ob_complete_strata.pdf](#)

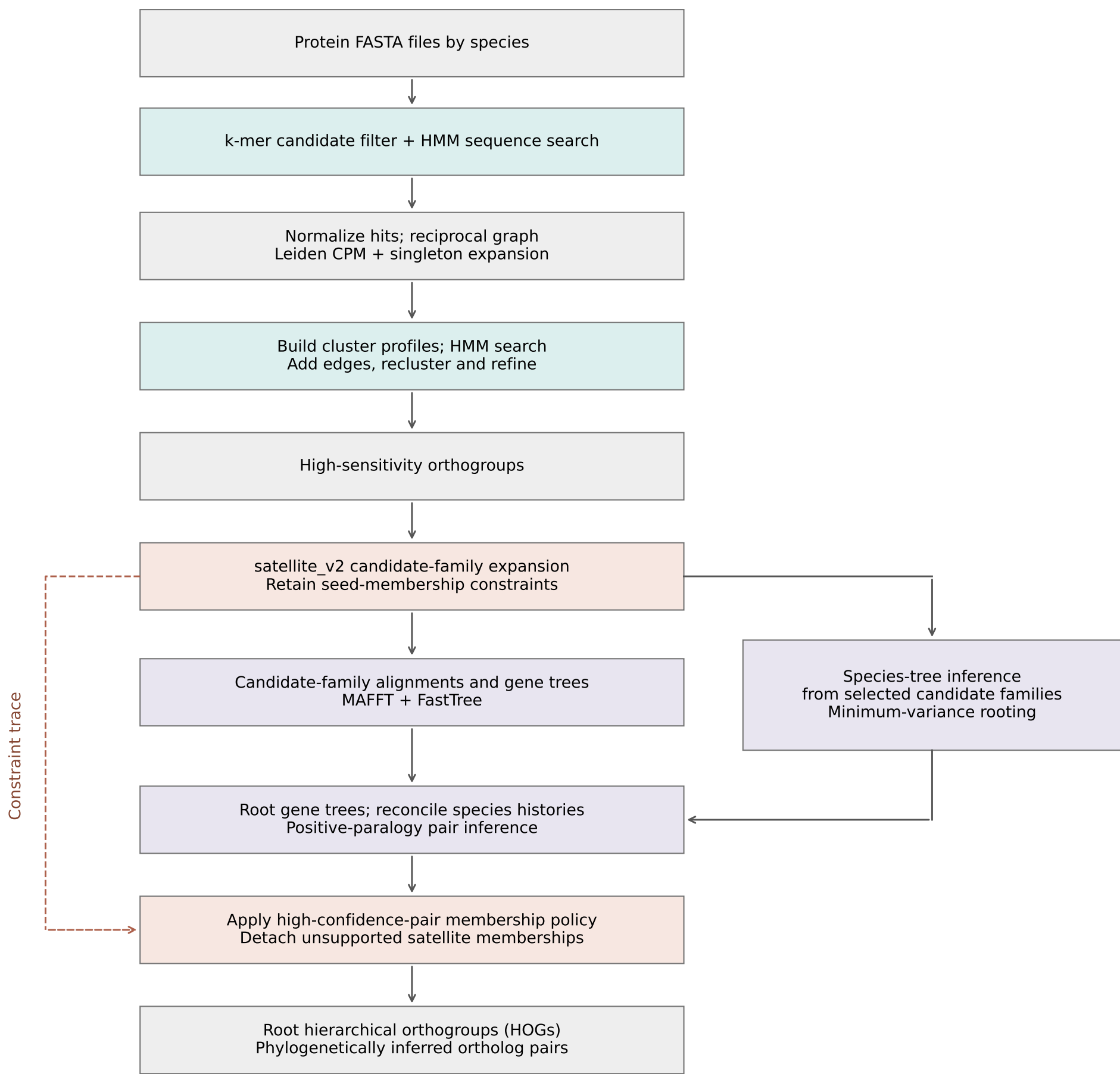
A16. Completed shared-host resource panel

All 27 attempts are reviewed: 25 measured, 24 eligible; six cells have three eligible repeats. Three cells lack a third eligible repeat and remain unavailable. Matched CPU/RAM limits do not remove unknown, potentially tool-dependent contention; no isolated efficiency ranking follows.

Source: [benchmark_tools/results/threadripper_shared_resource_figure_20261004_v26/shared_threadripper_resources.pdf](#)

OrthoHMM: frozen publication workflow

HMM-centered grouping followed by candidate expansion and inferred-tree refinement

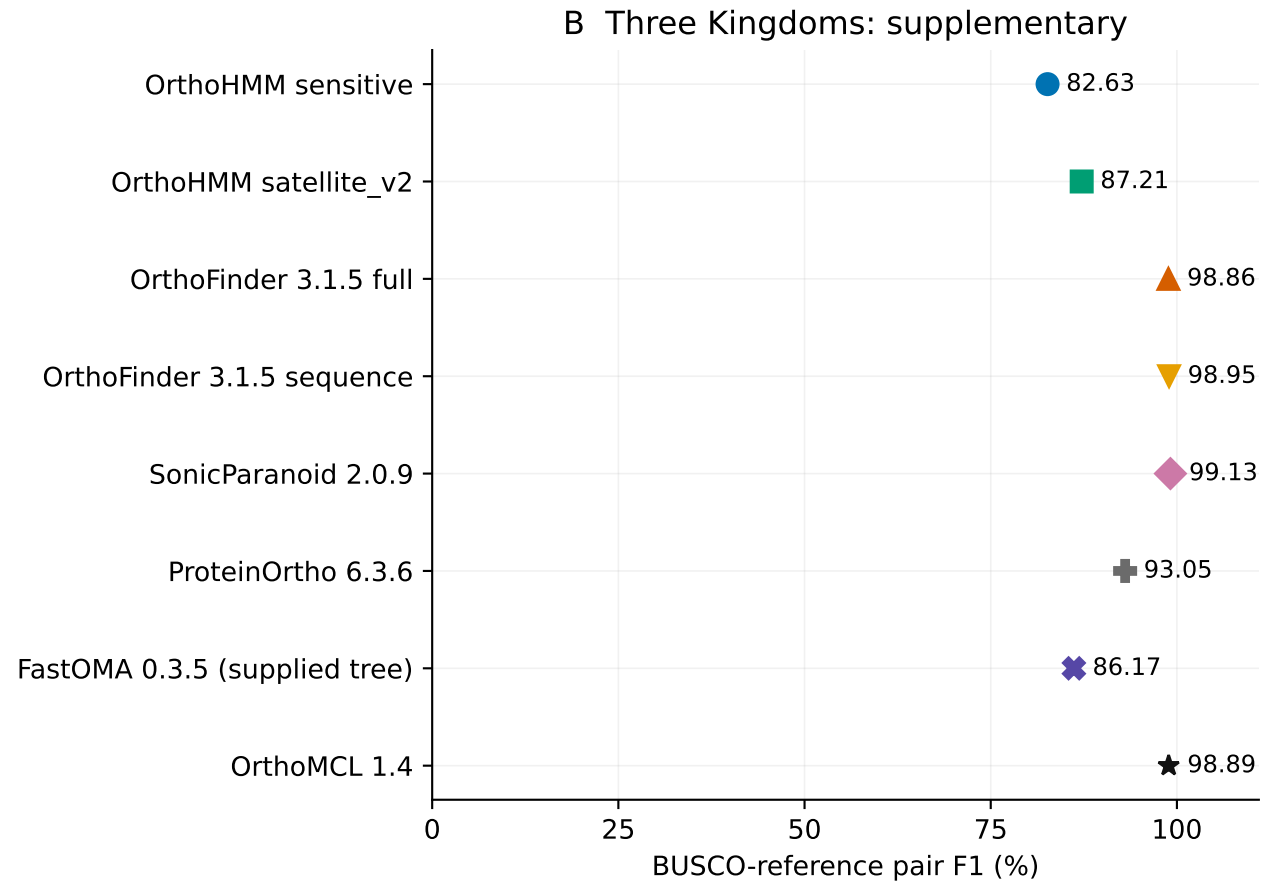
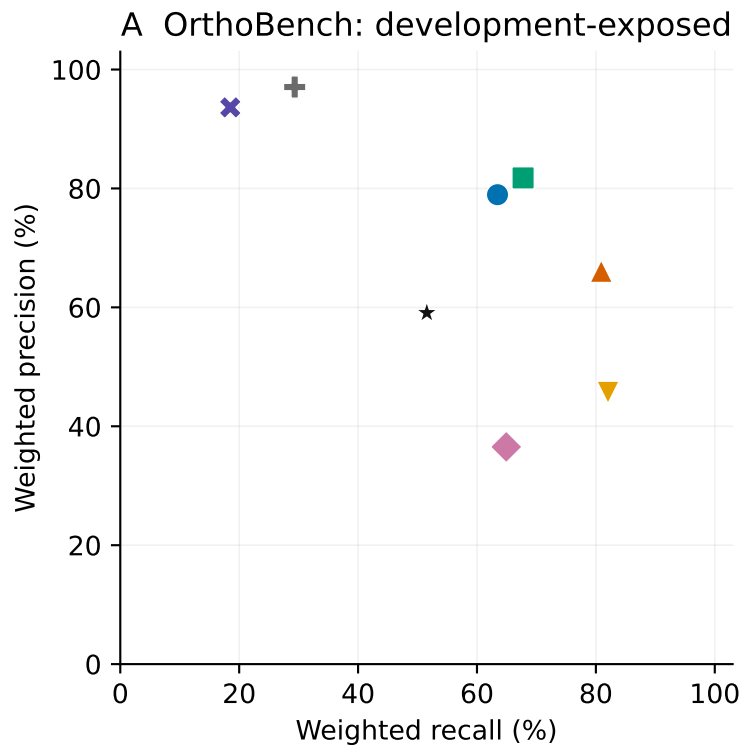


Schematic of the full satellite_v2 configuration; high-sensitivity output stops at orthogroups.

Small families may bypass tree inference. Exact-input checkpoints change cost, not the intended analysis.

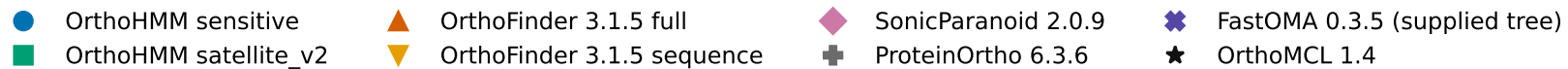
Output levels are distinct: group co-membership is not itself a phylogenetic orthology assignment.

Current retained group recovery | Eight methods

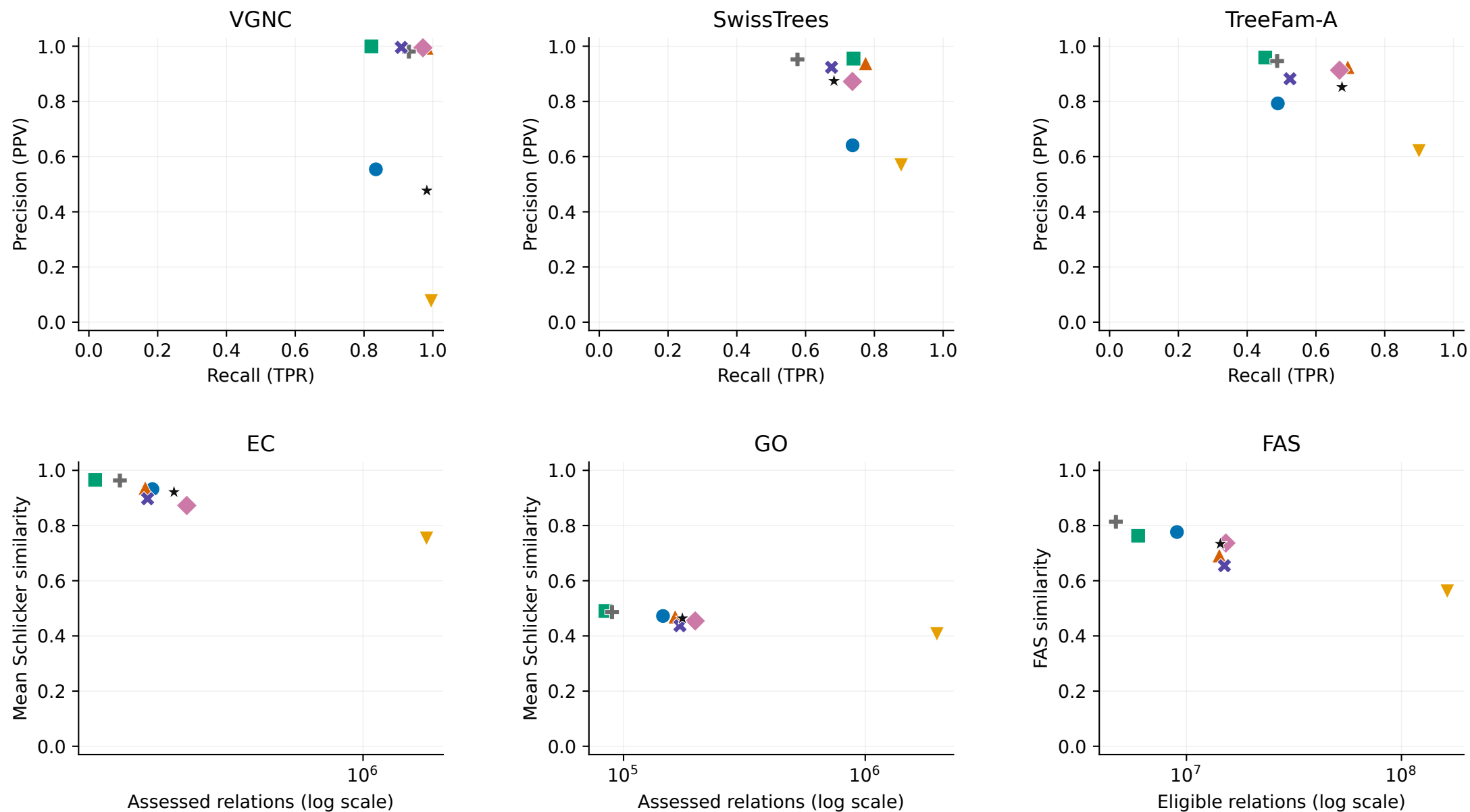


BUSCO-only pairs include within-species pairs and exclude non-reference-gene errors.

Descriptive points only; historical input-consumption gaps remain. OrthoFinder sequence is an MCL checkpoint.



Corrected QfO | Eight methods | Development-exposed

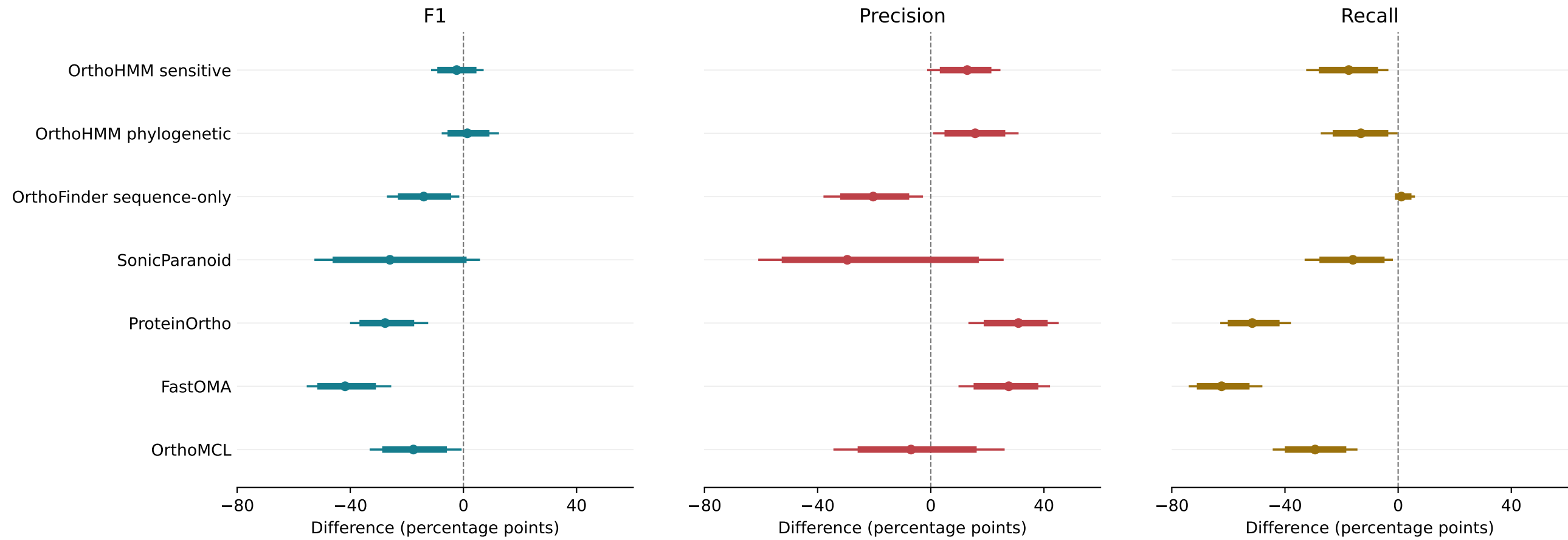


No cross-endpoint error bars: uncertainty definitions differ.
FAS x-axis is the reported eligible count, not the sampled count.

- OrthoHMM sensitive
- OrthoHMM satellite_v2
- OrthoFinder 3.1.5 full
- OrthoFinder 3.1.5 sequence
- SonicParanoid 2.0.9
- ProteinOrtho 6.3.6
- FastOMA 0.3.5 (supplied tree)
- OrthoMCL 1.4

OrthoBench: seven methods minus full OrthoFinder 3.1.5

70 RefOGs; 100,000 paired draws. Thick: nominal 95% interval. Thin: 21-endpoint adjusted interval.

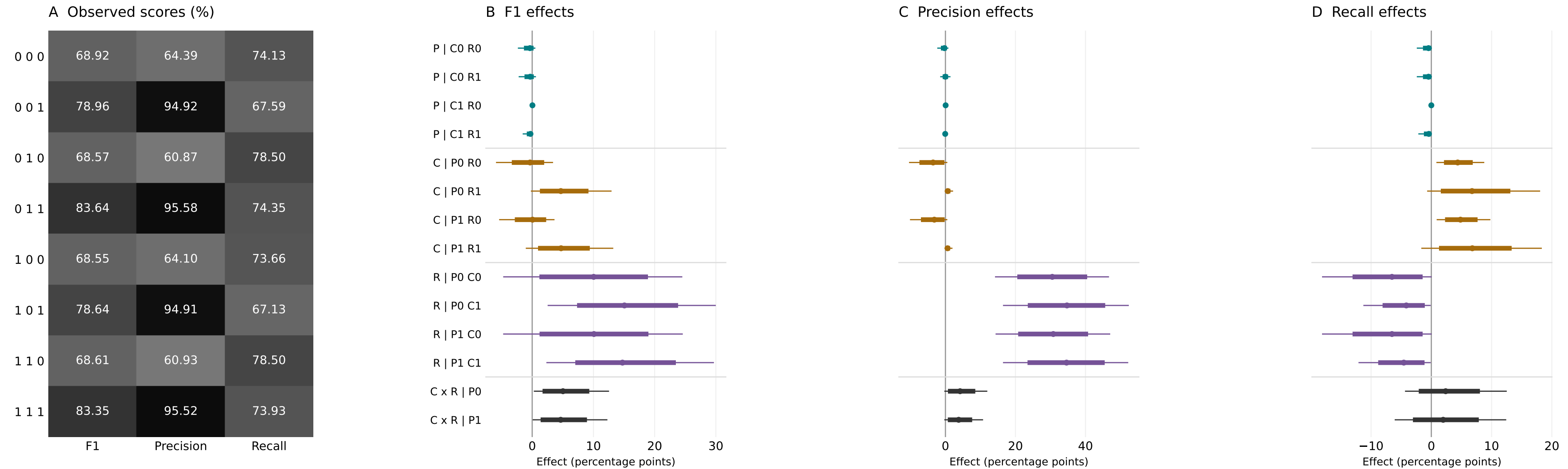


Development-exposed, conditional on family exchangeability. Intervals are approximate; inclusion of zero does not establish equivalence. Sequence-only OrthoFinder is a pre-phylogeny diagnostic. Fixed method order; no new inference, resampling or significance tests.

QfO SwissTrees: HMM refinement, candidate expansion and reconciliation

Corrected-release inputs (984,137 genes) | 18 reference families | development-exposed analysis

P = profile refinement; C = candidate expansion; R = reconciliation. Cells show P C R (0: off, 1: on).



Rows: P (teal), C (ochre), R (purple): on minus off, holding other factors fixed. Final rows: C x R interactions (gray).

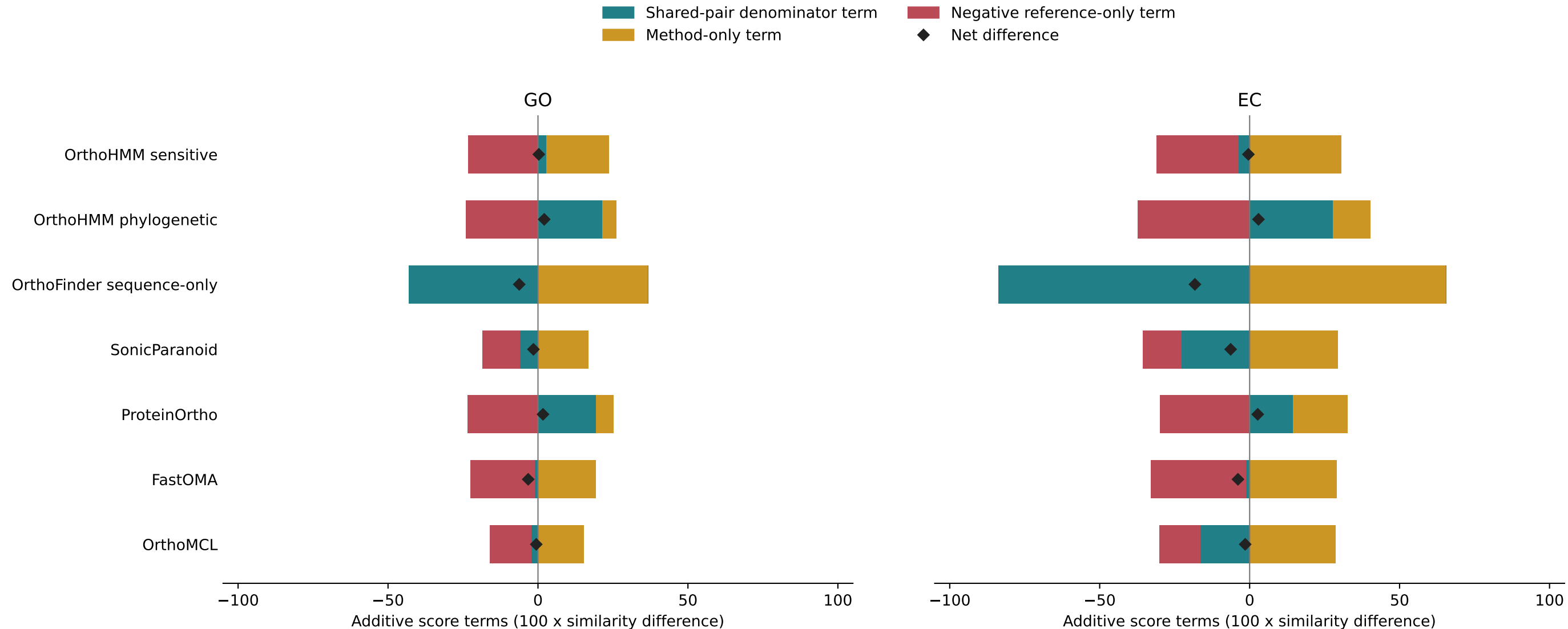
Thick: nominal 95% CI. Thin: Bonferroni CI across 42 endpoints. 100,000 paired family draws; seed 20260922.

F1 is the harmonic mean of macro precision and recall, recomputed per draw. 4/14 adjusted F1 intervals exclude zero.

Profile-off retains initial HMM search. R changes group-derived pairs to native inferred pairs, not only group splitting.

No independent validation, selection adjustment, or intervals for other QfO endpoints or the secondary mean.

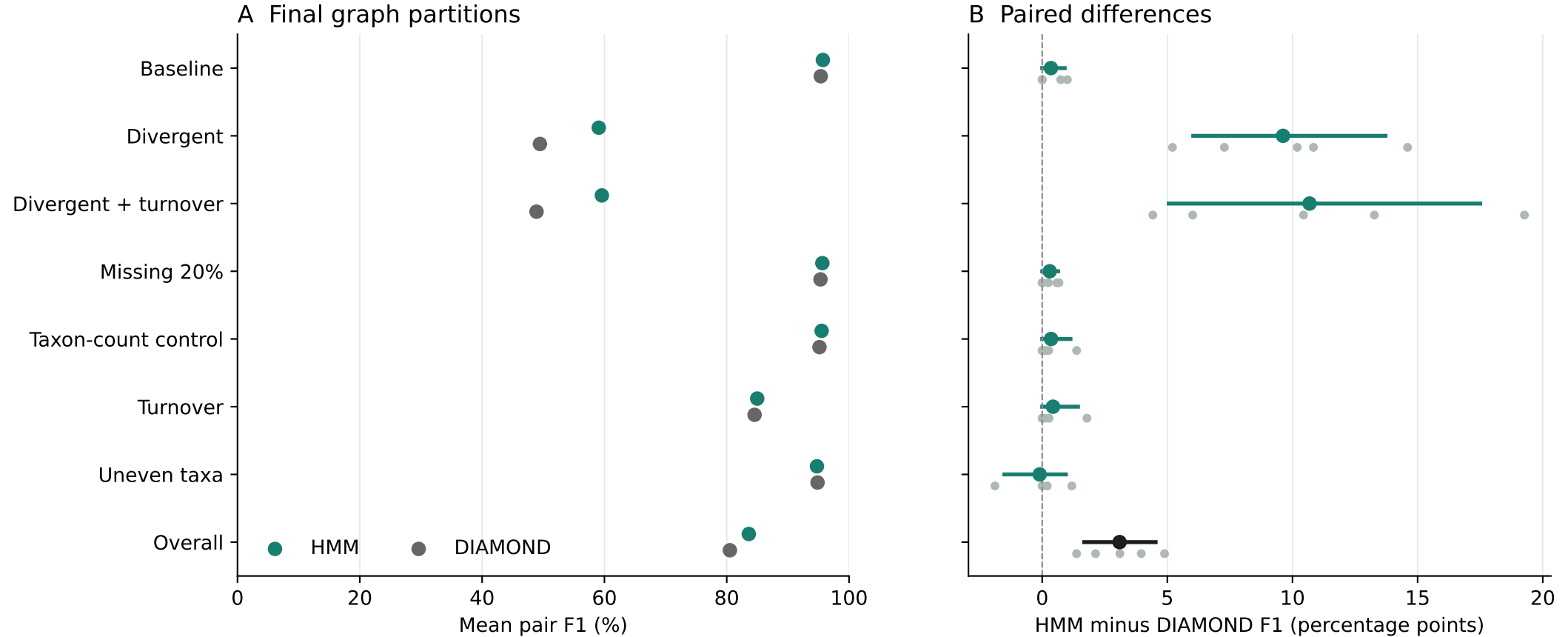
QfO: eligible-pair composition of GO/EC score differences



Seven methods minus full OrthoFinder 3.1.5; shared pairs have identical serialized scores. Not confidence intervals or causal effects.
Different prediction semantics and eligibility remain. Intersection conditioning changes the endpoint; native scores are not replaced.

Matched-recall initial-search control

35 reporting datasets; five seed blocks across seven conditions
Same production graph/refinement; profile expansion and phylogeny off

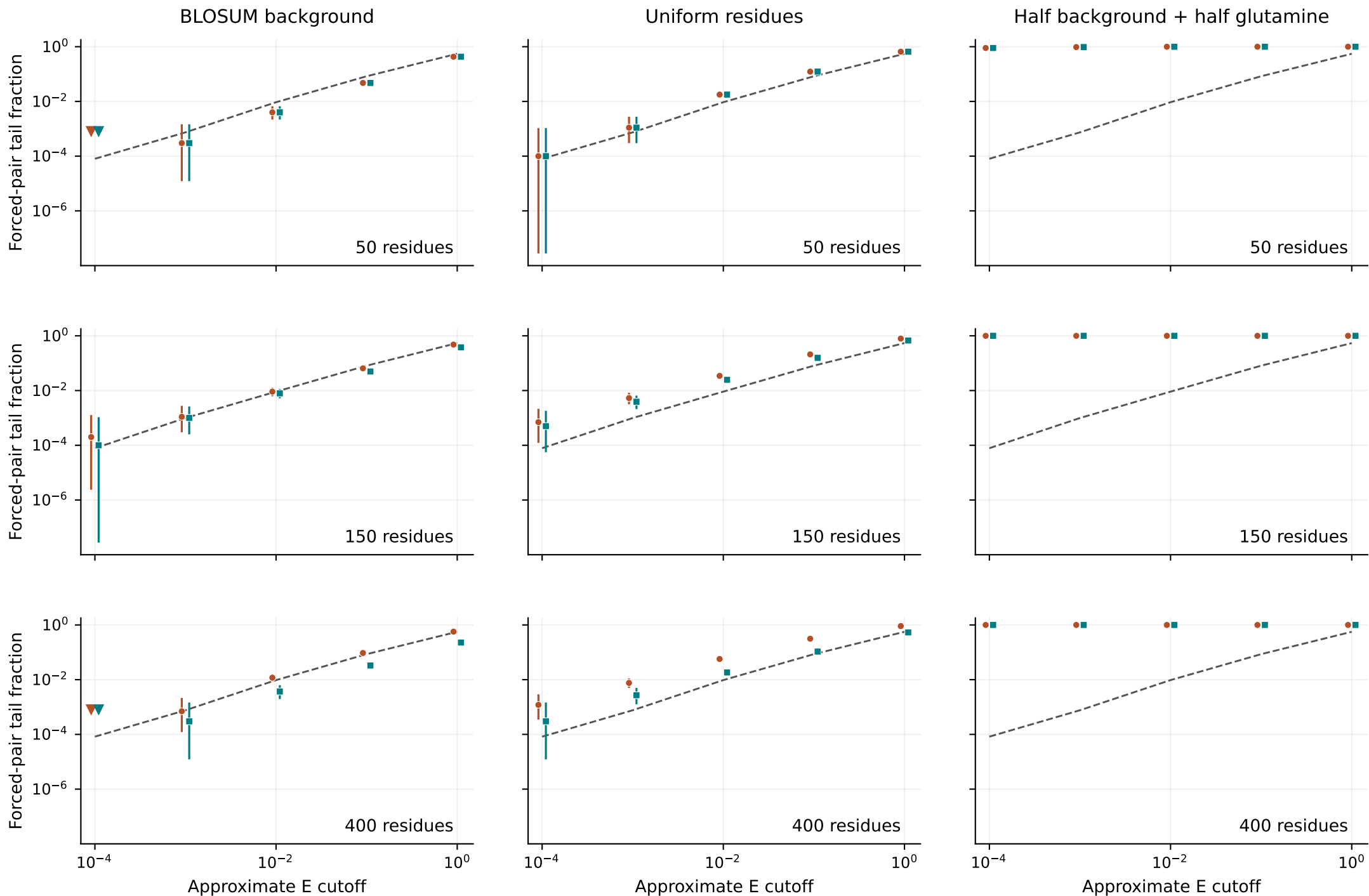


Intervals: paired seed-block percentile bootstrap, Bonferroni-adjusted for eight F1 contrasts.
Small dots: five seed effects. Five blocks give limited tail resolution and approximate coverage.
Development-exposed simulations; not real-data equivalence or full-pipeline superiority.

Synthetic null-score tails

Frozen recurrence and significance formula; no parameter fitting

● Full matrix ■ Width 64 - - - Poisson-model reference ▼ Zero-hit upper bound



10,000 independent pairs per panel; whiskers: exact intervals adjusted over 90 planned endpoints.

Forced scoring only, with one target per query. No prefilter, grouping or orthology inference.

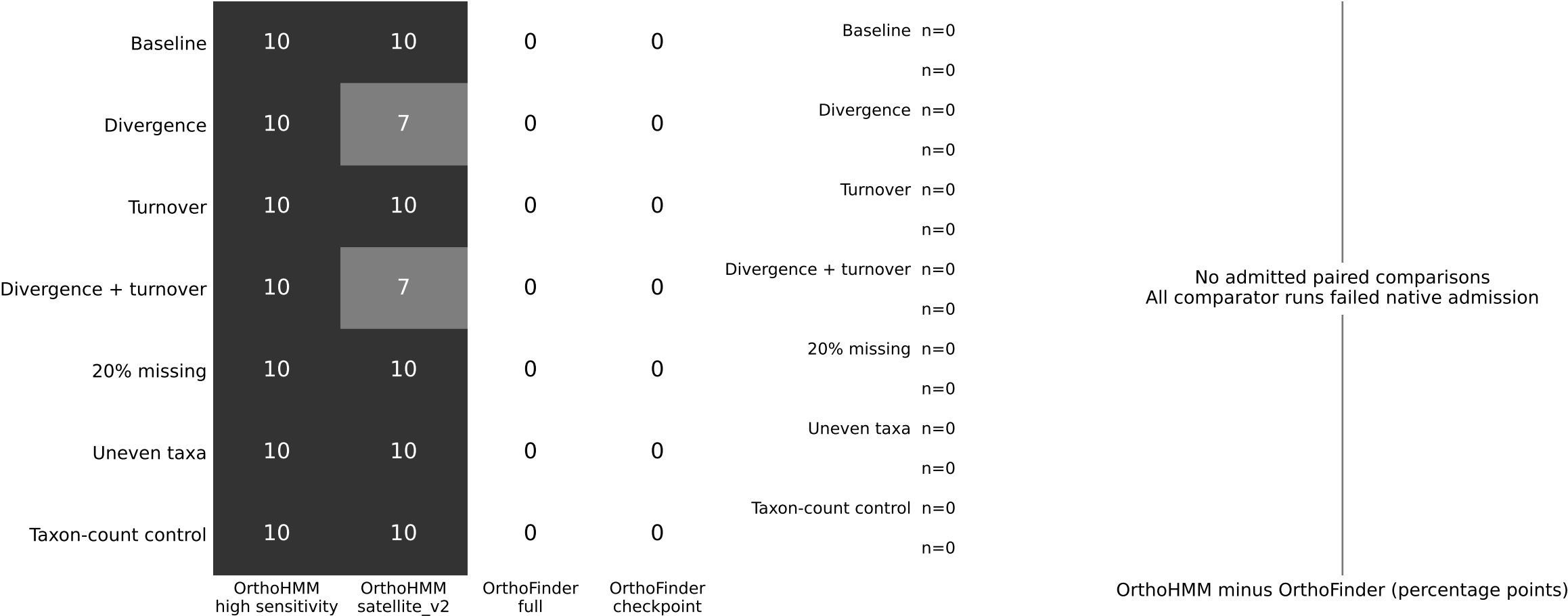
Zero-hit triangles mark interval upper bounds, not positive observed fractions; band glyphs are offset slightly.

Corrected simulations: Fixed-length stress panel

OrthoHMM high sensitivity OrthoHMM satellite_v2

A Admitted datasets (out of 10)

B Paired F1 difference versus full OrthoFinder



Thick lines: nominal 95% CI. Thin lines: Bonferroni-14 CI. Bootstrap unit: independent simulation seed.

Comparisons condition on joint success; failed runs receive no imputed score. Panels are not pooled.

Ten seeds limit interval precision. The sequence checkpoint requires a valid full parent run.

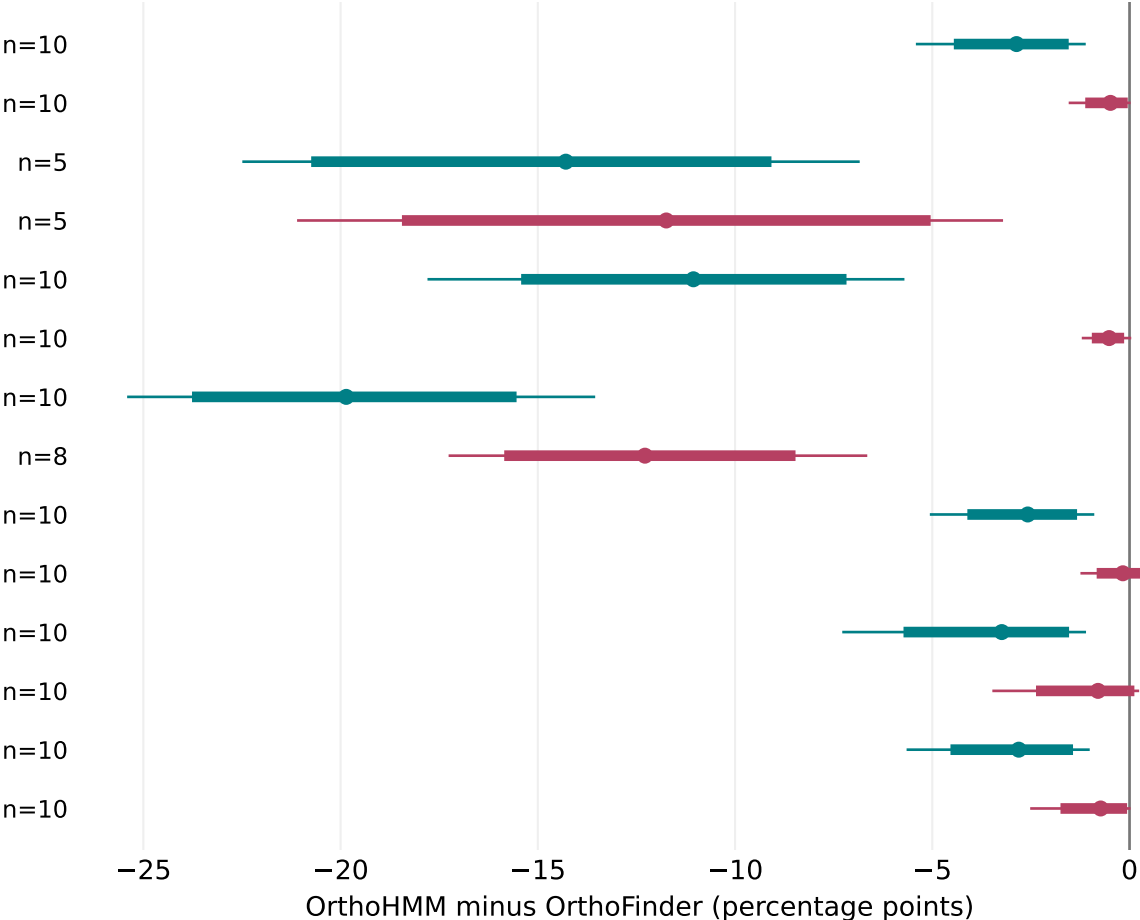
Corrected simulations: Variable family lengths

OrthoHMM high sensitivity OrthoHMM satellite_v2

A Admitted datasets (out of 10)

Baseline	10	10	10	10
Divergence	10	9	5	5
Turnover	10	10	10	10
Divergence + turnover	10	8	10	10
20% missing	10	10	10	10
Uneven taxa	10	10	10	10
Taxon-count control	10	10	10	10
	OrthoHMM high sensitivity	OrthoHMM satellite_v2	OrthoFinder full	OrthoFinder checkpoint

B Paired F1 difference versus full OrthoFinder



Thick lines: nominal 95% CI. Thin lines: Bonferroni-14 CI. Bootstrap unit: independent simulation seed.

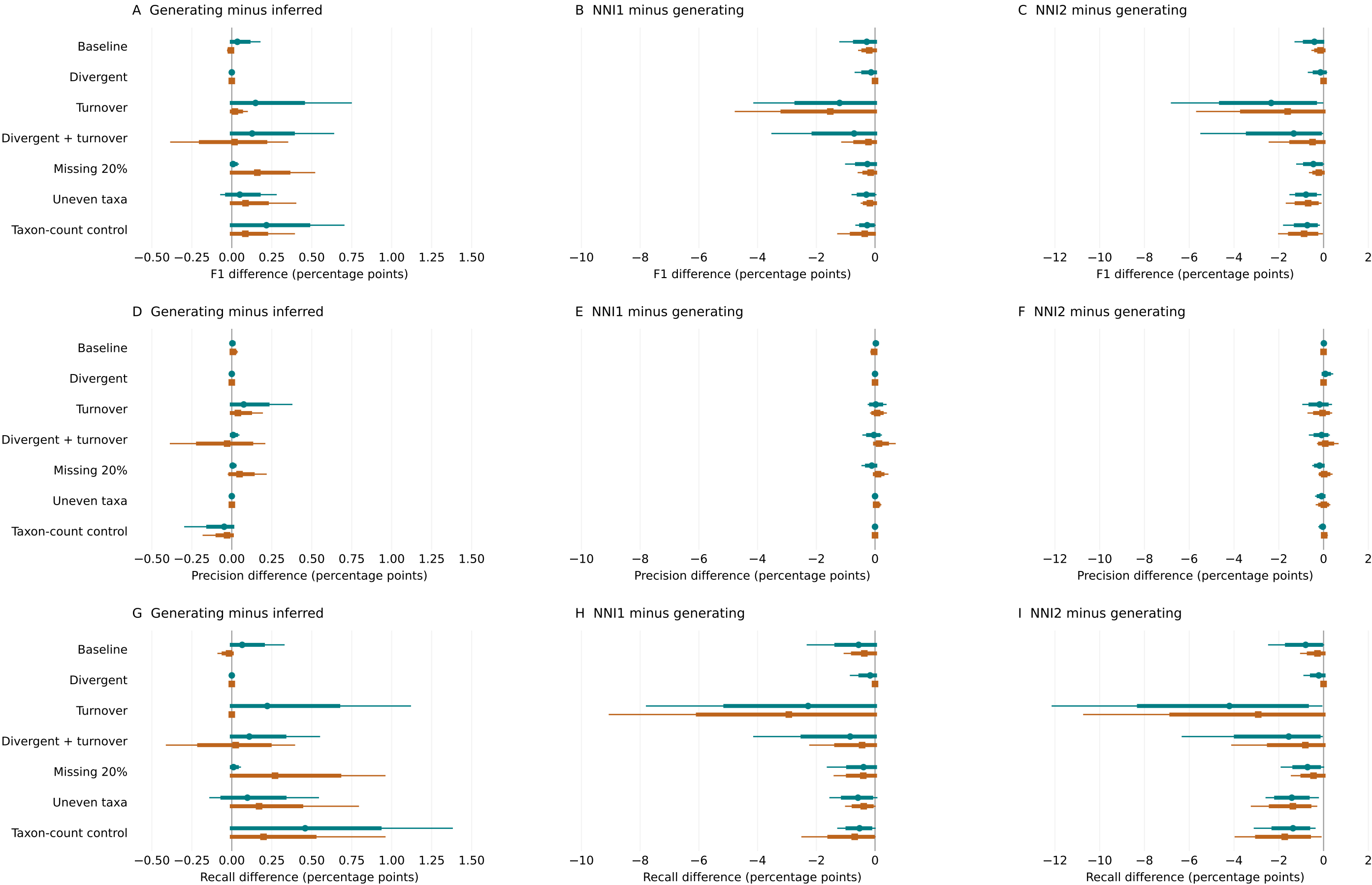
Comparisons condition on joint success; failed runs receive no imputed score. Panels are not pooled.

Ten seeds limit interval precision. The sequence checkpoint requires a valid full parent run.

Simulation sensitivity to supplied species trees

70 datasets | 560 planned method/tree outcomes | 23 failures retained | 126 exploratory endpoints

● OrthoHMM satellite_v2 ■ OrthoFinder full 3.1.5



Dots/squares: paired-seed means. Thick: nominal 95% CI. Thin: Bonferroni CI across all 126 endpoints.

20,000 paired-seed draws; seed 20260918. Available paired counts: 5, 8, 9, 10. Failures are not assigned zero accuracy.

Generating tree is an oracle diagnostic. Retained upstream differences and per-contrast exclusions are reported in the accompanying audit.

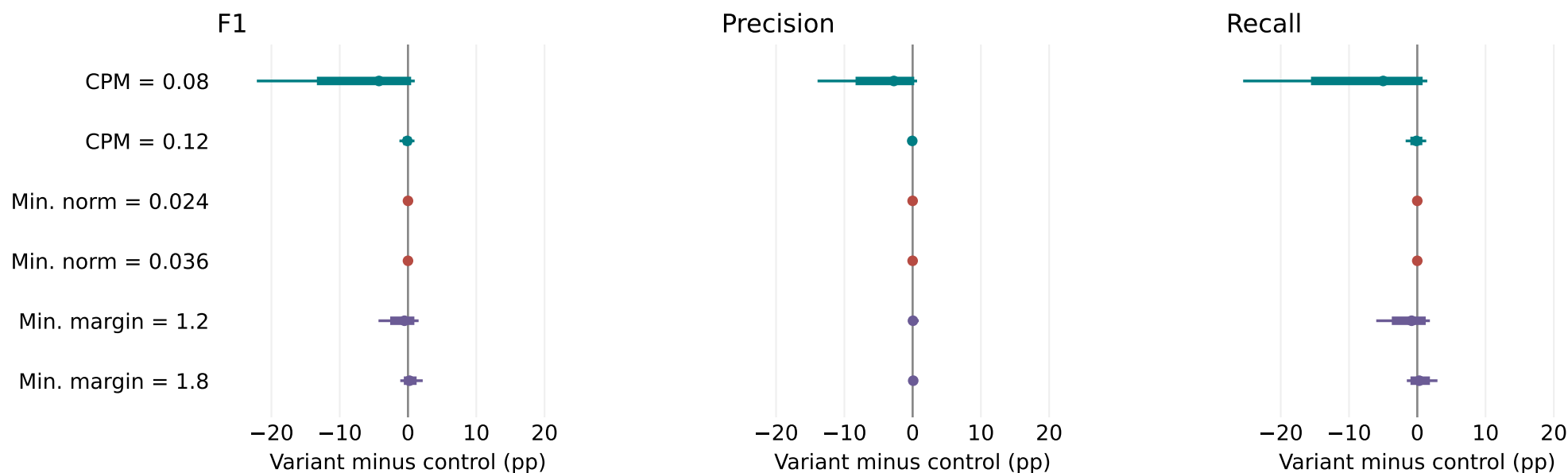
Development-exposed simplified simulations; small seed counts and approximate bootstrap tails limit generalization and simultaneous coverage.

QfO SwissTrees: limited parameter sensitivity

Six one-at-a-time changes of 20% | 18 reference families | development-exposed

6/6 contrasts estimated; unavailable arms are not zero. No default selection.

Configuration	F1 (%)	Precision (%)	Recall (%)
Frozen control	83.351	95.518	73.934
CPM = 0.08	79.101	92.771	68.943
CPM = 0.12	83.253	95.461	73.813
Min. norm = 0.024	83.351	95.518	73.934
Min. norm = 0.036	83.351	95.518	73.934
Min. margin = 1.2	82.839	95.569	73.102
Min. margin = 1.8	83.565	95.603	74.219



Thick: paired 95% CI. Thin: Bonferroni CI over all 18 planned endpoints; 100,000 draws, seed 20260925.

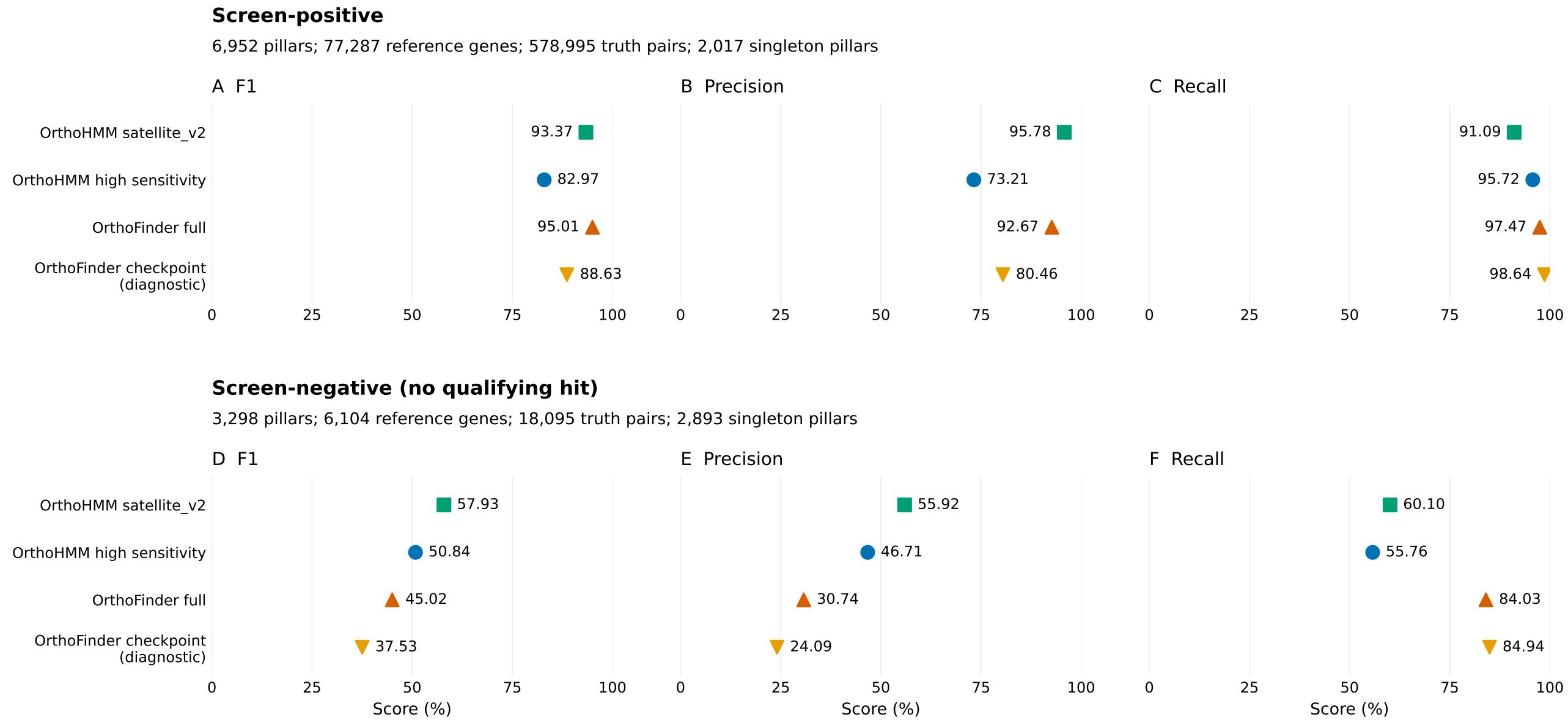
F1 is recomputed from macro precision and recall per draw; missing contrasts retain their places.

No independent confirmation, selection adjustment, OrthoFinder superiority test or controlled timing is shown.

Intervals including zero do not establish equivalence. No other-endpoint or secondary-mean uncertainty is shown.

YGOB overlap screen: group-recovery trade-offs

Secondary descriptive analysis | all four frozen methods | no new confidence intervals



No-hit status does not establish family independence or absence of remote homology.

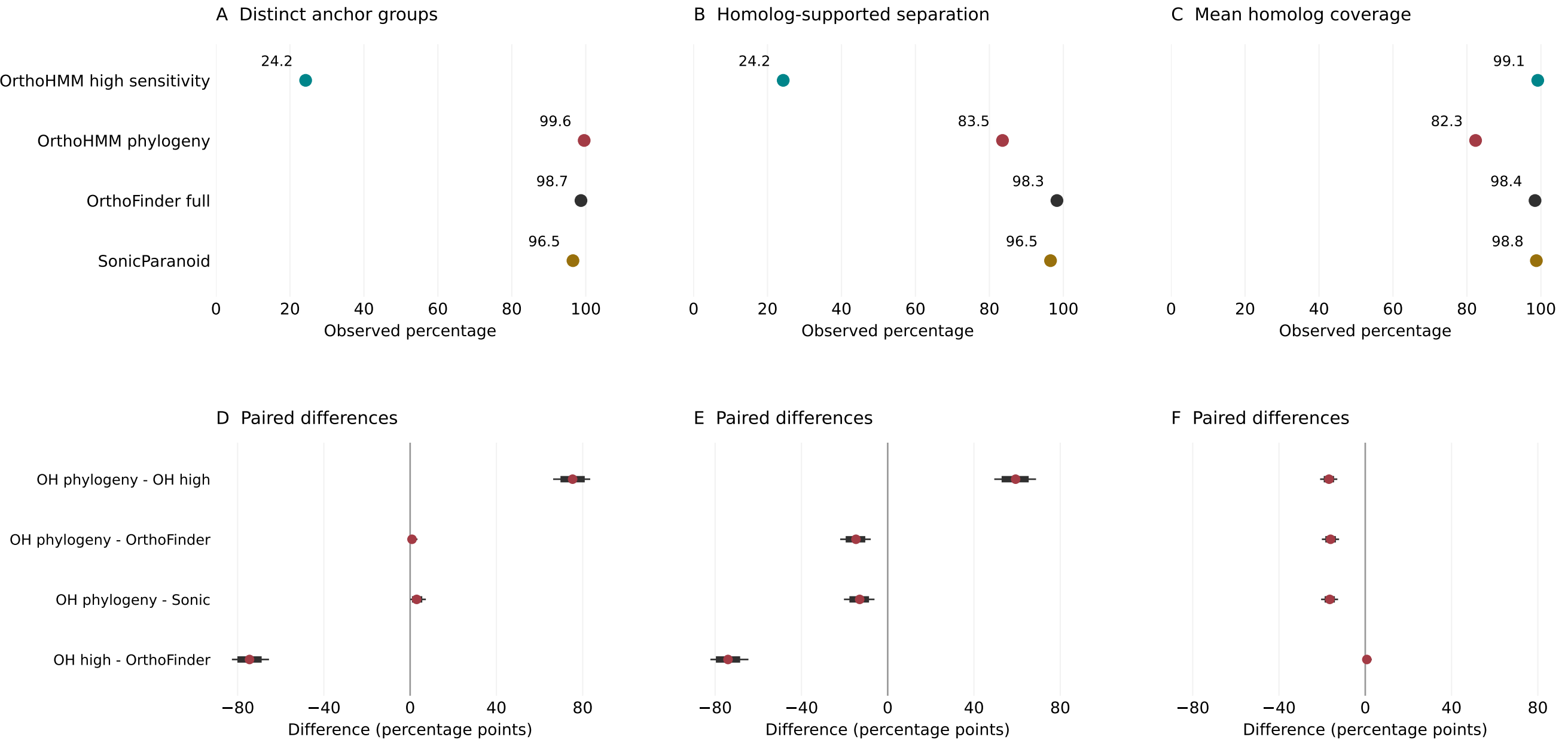
Original cross-pillar false-positive allocations retained; predictions were not subset-rescored.

Curated group co-membership includes within-species pairs; these are not resolved pairwise orthology scores.

Whole-genome-duplicate application

Development-exposed Saccharomyces data | all panels use the same 231 shared-pillar pairs

Homolog support is not copy-specific orthology truth. Coverage alone can reward merged paralogs.



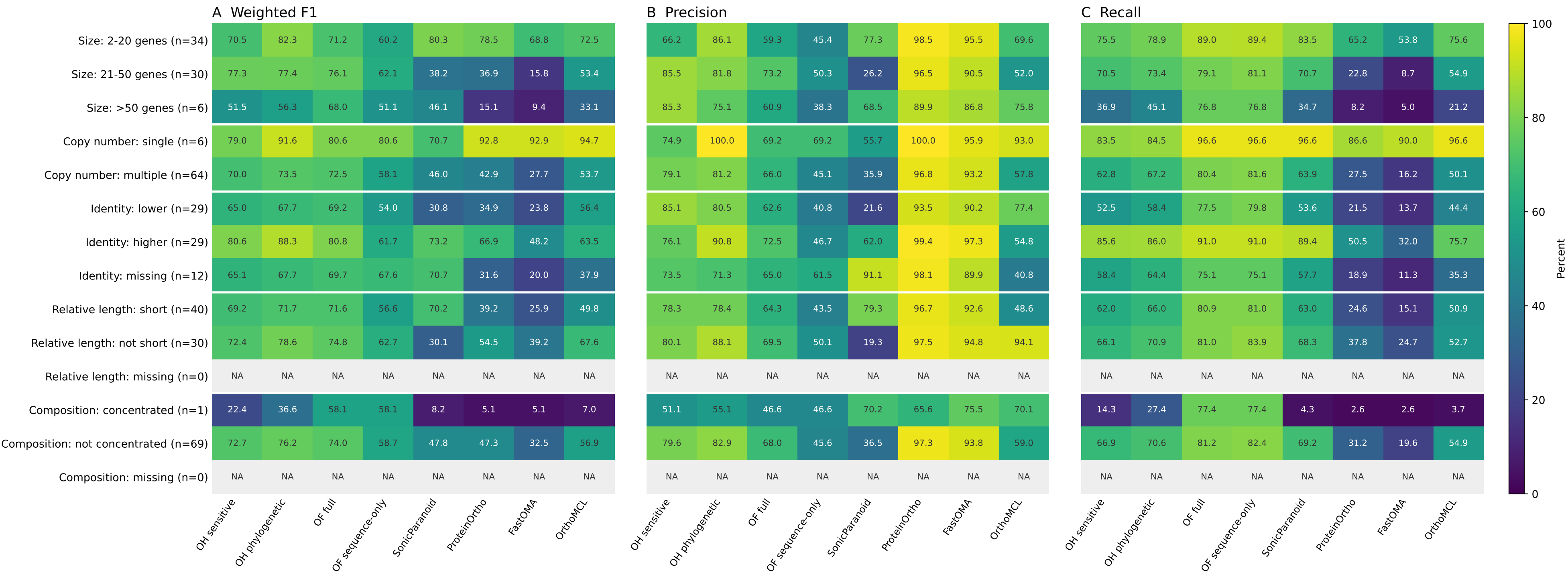
Thick: nominal 95% interval. Thin: Bonferroni interval across 12 contrasts. 20,000 paired pillar resamples; seed 20260920.

The OrthoHMM configurations differ in more than reconciliation; their contrast is not a pure phylogeny ablation.

All 240 experimental pairs, exclusions, the diagnostic MCL checkpoint and individual memberships remain in the supplement.

OrthoBench: all-method descriptive strata

Eight retained methods | 14 frozen bins | 70 reference families | fixed method order



Development-exposed descriptive results; no new confidence intervals or significance tests. Empty bins are NA, not zero.

Statistics recomputed from family-size-weighted TP/FP/FN, not mean family F1. Bins overlap across dimensions; n counts reference families.

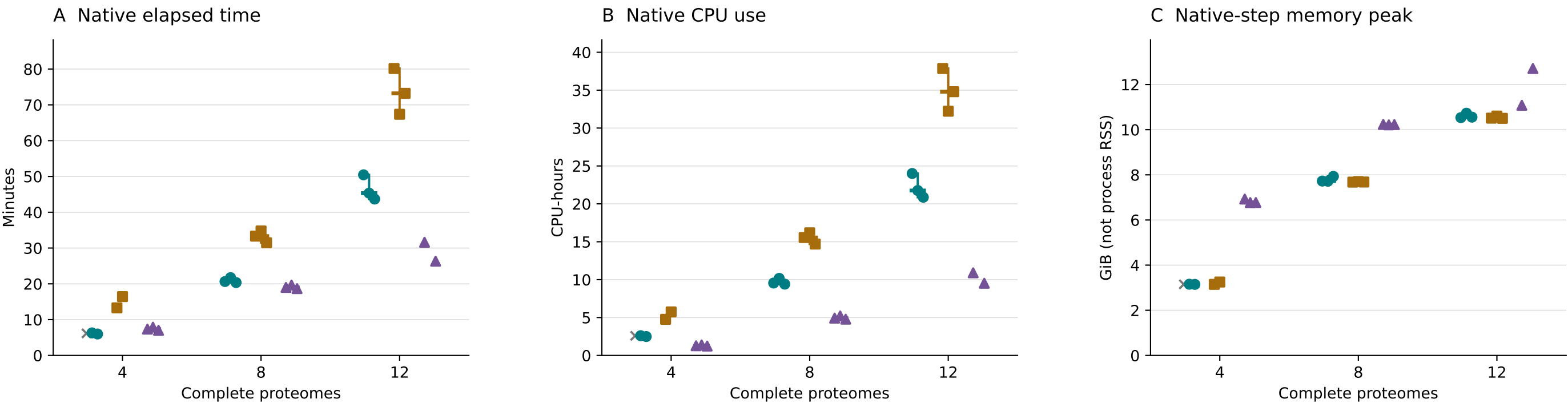
OH = OrthoHMM; OF = OrthoFinder 3.1.5. Descriptors are not validated domain, fragmentation or duplication-history annotations.

Threadripper shared-host resource observations

27/27 attempts reviewed | 25 with measured resources | 24 eligible | 2 pre-native aborts

32 native CPUs and 128 GiB per run; shared host, not isolated timing

OrthoHMM high sensitivity OrthoHMM inferred phylogeny OrthoFinder 3.1.5 full × Excluded measured attempt



Symbols: measured attempts. Gray crosses: excluded measurements, never pooled into eligible summaries.

Pre-native aborts have unavailable endpoints and no plotted symbol; they are not zero or unattempted runs.

Median ticks/range bars require three eligible repeats; ranges are not confidence intervals.

Contention distortion is unknown and potentially method dependent. No causal speedup or overhead subtraction.

CPU: task-subtree bracket including wrapper. Peak: native-step lifetime including launcher, not algorithm RSS.

One nested taxon series: 73,266 / 165,168 / 251,378 proteins; taxon composition and proteome count co-vary.

OrthoHMM high sensitivity | 4 proteomes: 2/3 eligible, 1 excluded; 8 proteomes: 3/3 eligible, 0 excluded; 12 proteomes: 3/3 eligible, 0 excluded

OrthoHMM inferred phylogeny | 4 proteomes: 2/3 eligible, 1 excluded; 8 proteomes: 3/3 eligible, 0 excluded; 12 proteomes: 3/3 eligible, 0 excluded

OrthoFinder 3.1.5 full | 4 proteomes: 3/3 eligible, 0 excluded; 8 proteomes: 3/3 eligible, 0 excluded; 12 proteomes: 2/3 eligible, 1 excluded